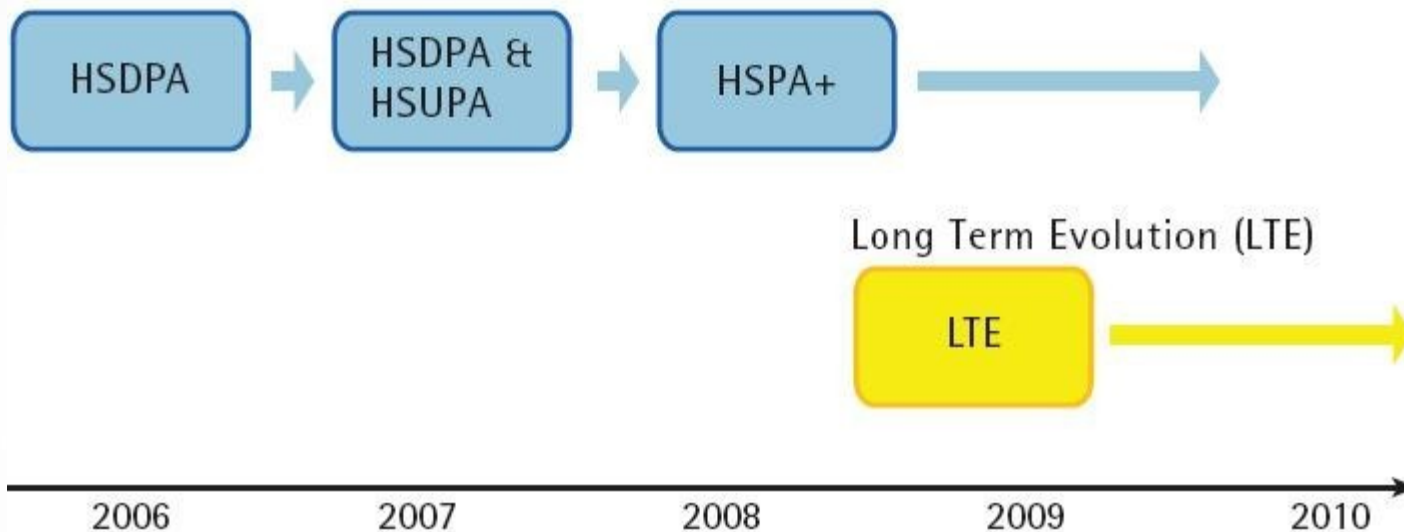




LTE(Long Term Evolution)

3G evolution

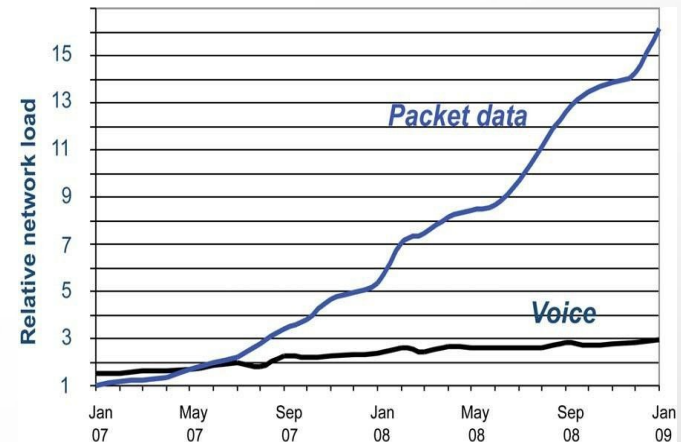
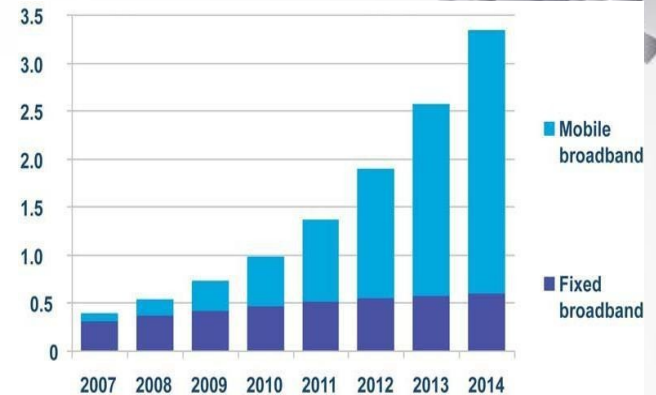
UMTS Evolution



Evolution timeframe for planned 3GPP systems.

Why LTE.....?

- **Need for PS optimized system**
 - Evolve UMTS towards packet only system
- **Need for higher data rates**
 - Can be based on transfer speed of Downlink Packet Access or new air interface defined by 3GPP LTE
- **Less processor load cost**
 - Less number of transitions between different states will lead definitely to less processor load
- **Need for high quality of services**
 - Use of licensed frequencies to guarantee quality of services
 - Always-on experience (reduce control plane latency significantly)
 - Reduce round trip delay (→ 3GPPLTE)
- **Need for cheaper infrastructure**
 - Simplify architecture, reduce number



LTE Defined Data Rates



- Downlink
 - 100Mbps theoretical
- Uplink
 - 50Mbps theoretical

LTE duplexing and accessing



- Duplexing Methods

- FDD

- UL and DL can reach the peak traffic simultaneously

- TDD

- UL and DL can not reach the peak traffic simultaneously

- Accessing techniques

- OFDMA for the DL

- SC-FDMA for the UL

Modulation and coding



- Adaptive Modulation and Coding
 - DL Modulations: QPSK, 16-QAM and 64-QAM modulation
 - UL Modulations: QPSK and 16-QAM

LTE System Performance targets(1- 2)

- **User throughput**
 - 95% from the users will take average throughput
 - 5% will be little bit smaller than the average
- **Spectrum efficiency**
 - high spectrum efficiency
- **Coverage**
 - 5 Km with high throughput
 - 30 Km with low throughput
 - 100Km with very low throughput



LTE System Performance targets(2-2)

- **Mobility**

- 0-15 km/ hour the more better subscriber behavior.
- 120 km/ hour the accepted behavior.
- 350 km/ hour very low data rate and data throughput.

- **Enhanced MBMS(Multimedia Broadcast Multicast Service)**

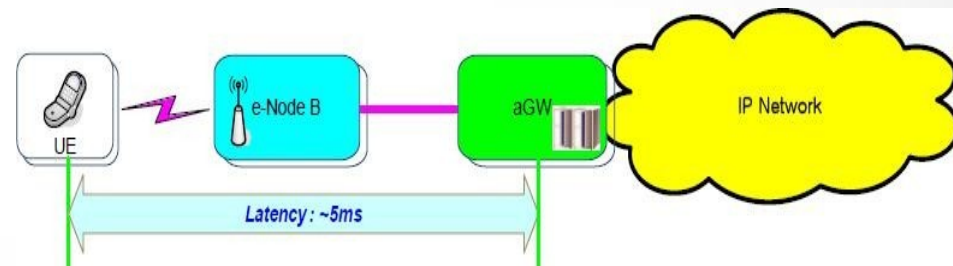
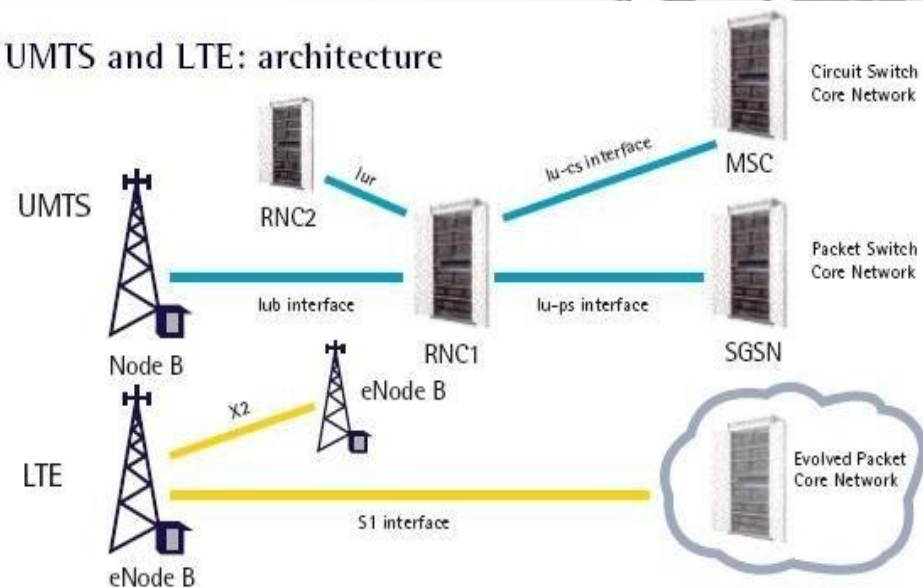
- Up to 16 multimedia channels per just one carrier



Architecture and Migration

- LTE RAN agreed on the following
 - Packet bearer support
 - Reduce the number of the new interfaces
 - NO RNC
 - NO CS-CN
 - Reduce the single point of failure
 - NO RNC
 - Separate the treatment of different types of traffic (O&M, Con and Data) to utilize the BW
- Reduce the variable delay and Jitter (TCP/ IP)
- Agreed QOS between Transmitting end and receiving end
- Macro diversity
- MIMO and Tx diversity techniques used

UMTS and LTE: architecture



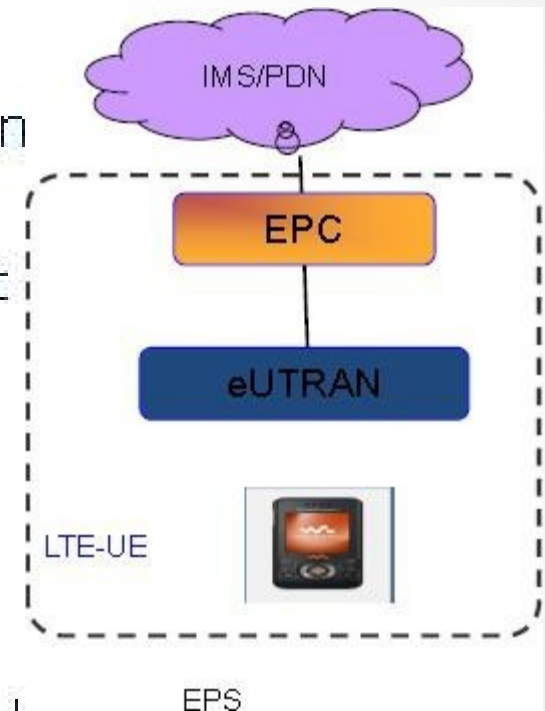
Complexity



- Easy design
- Less complex
- No redundant feature
- Minimize Cost and maintain system performance
 - Low complexity
 - Low power consumption

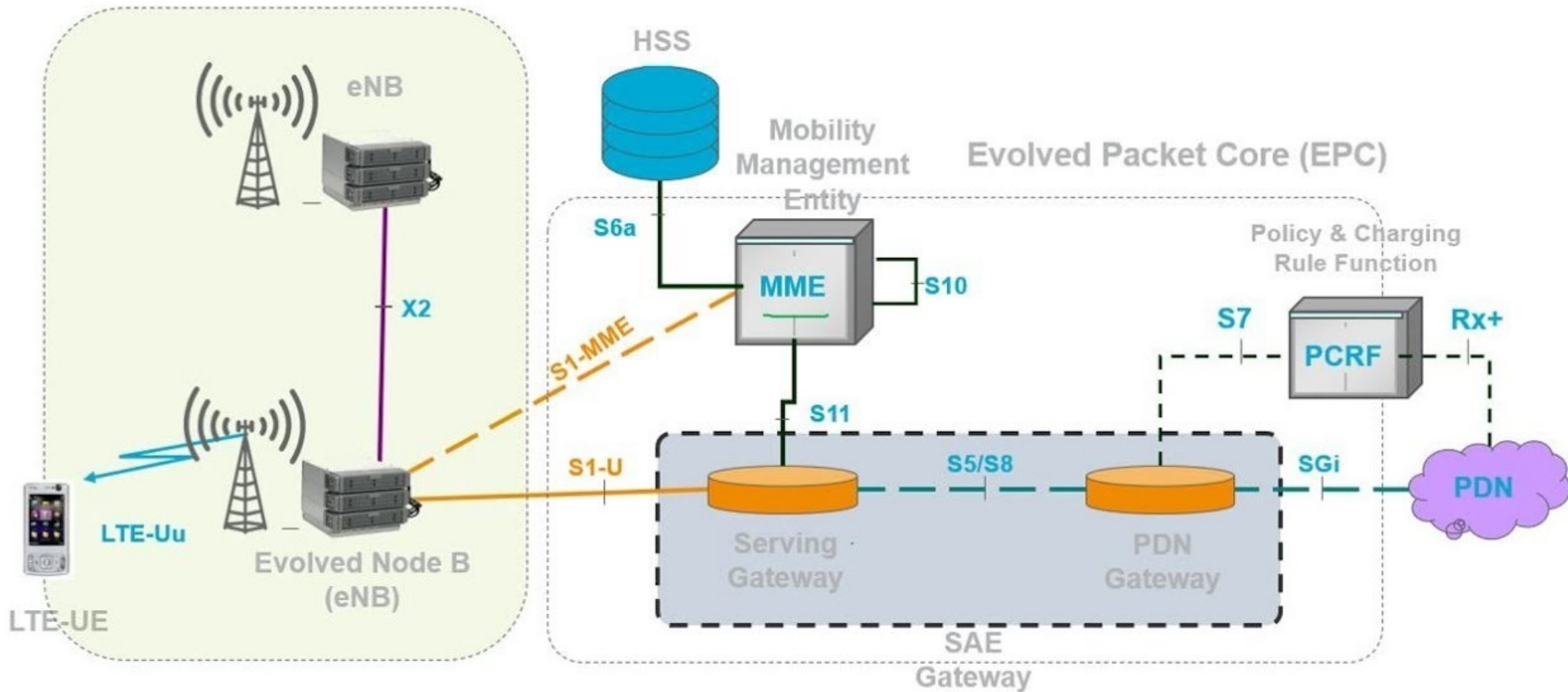
3GPP-LTE Architecture High level

- LTE/SAE architecture is driven by the goal to optimize the system for packet data transfer.
- No circuit switched components
- New approach in the inter-connection between radio access network and core network
- The EPS architecture is made up of an EPC (Packet Core Network, also referred as SAE) and an eUTRAN Radio Access Network (also referred as LTE)
- The CN provides access to external packet IP networks and performs a number of CN related functions (e.g. QoS, security, mobility and terminal context management) for idle (camped) and active terminals
- The RAN performs all radio interface related functions



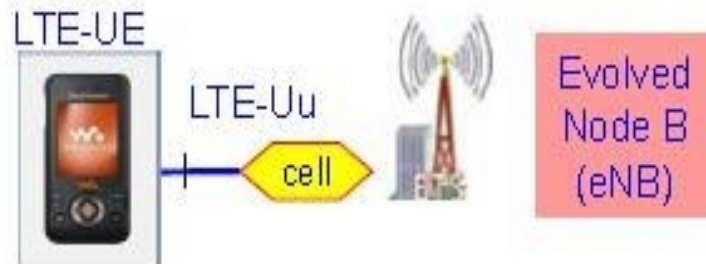
LTE Architecture

Evolved UTRAN (E-UTRAN)



Evolved UTRAN

Evolved Node B (eNB)



- It is the only network element defined as part of EUTRAN.
- It replaces the old Node B / RNC combination from 3G.
- It terminates the complete radio interface including physical layer.
- It provides all radio management functions
- An eNB can handle several cells.
- To enable efficient inter-cell radio management for cells not attached to the same eNB, there is an inter-eNB interface X2 specified. It will allow to coordinate inter-eNB handovers without direct involvement of EPC during this process.

eNB Functions

Inter-cell RRM: HO, load balancing between cells

Radio Bearer Control: setup, modifications and release of Radio Resources

Connection Mgt. Control: UE State Mgmt. MME-UE Connection

Radio Admission Control

eNode B Measurements Collection and evaluation

Dynamic Resource Allocation (Scheduler)

IP Header Compression/ de-compression

Access Layer Security: ciphering and integrity protection on the radio interface

MME Selection at Attach of the UE

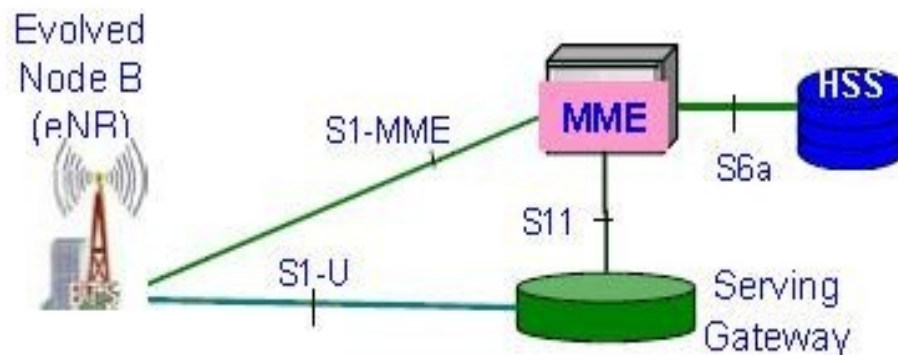
User Data Routing to the SAE GW.

Transmission of Paging Message coming from MME

Transmission of Broadcast Info (System info, MBMS)

EPC (1-5)

Mobility Management Entity (MME)



- It is a pure signaling entity inside the EPC.
- SAE uses tracking areas to track the position of idle UEs. The basic principle is identical to location or routing areas from 2G/3G.
- MME handles attaches and detaches to the SAE system, as well as tracking area updates
- Therefore it possesses an interface towards the HSS (home subscriber server) which stores the subscription relevant information and the currently assigned MME in its permanent data base.
- A second functionality of the MME is the signaling coordination to setup transport bearers (SAE bearers) through the EPC for a UE.
- MMEs can be interconnected via the S10 interface

MME Functions

Control plane NE in EPC

Non-Access-Stratum (NAS)
Security (Authentication,
integrity Protection)

Idle State Mobility Handling

Tracking Area updates

Subscriber attach/detach

Signalling coordination for
SAE Bearer Setup/Release

Radio Security Control

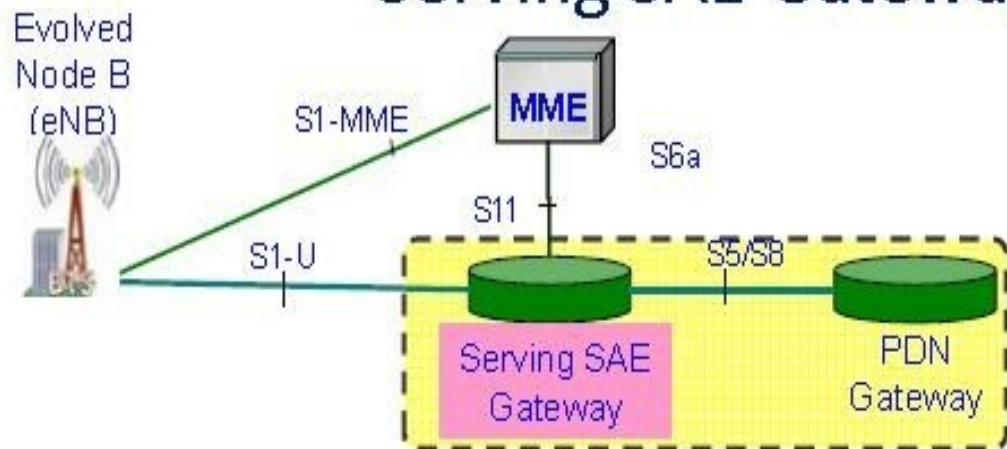
Trigger and distribution of
Paging Messages to eNB

Roaming Control (S6a interface
to HSS)

Inter-CN Node Signaling
(S10 interface), allows efficient
inter-MME tracking area updates
and attaches

EPC (2-5)

Serving SAE Gateway



- The serving gateway is a network element that manages the user data path (SAE bearers) within EPC.
- It therefore connects via the S1-U interface towards eNB and receives uplink packet data from here and transmits downlink packet data on it.
- Thus the serving gateway is some kind of distribution and packet data anchoring function within EPC.
- It relays the packet data within EPC via the S5/S8 interface to or from the PDN gateway.
- A serving gateway is controlled by one or more MMEs via S11 interface.

Serving Gateway Functions

Local mobility anchor point:
Switching the user plane path to a new eNB in case of Handover

Mobility anchoring for inter-3GPP mobility. This is sometimes referred to as the 3GPP Anchor function

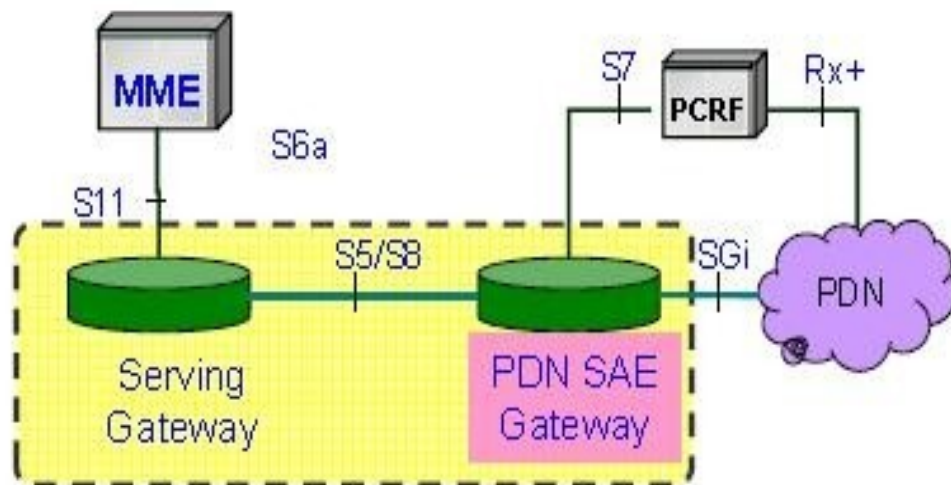
Idle Mode Packet Buffering and notification to MME

Packet Routing/Forwarding
between eNB, PDN GW and SGSN

Lawful Interception support

EPC (3-5)

Packet Data Network (PDN) SAE Gateway



PDN Gateway Functions

Mobility anchor for mobility between 3GPP access systems and non-3GPP access systems. This is sometimes referred to as the SAE Anchor function

Policy Enforcement (PCEF)

Per User based Packet Filtering (i.e. deep packet inspection)

Charging & Lawful Interception support

IP Address Allocation for UE

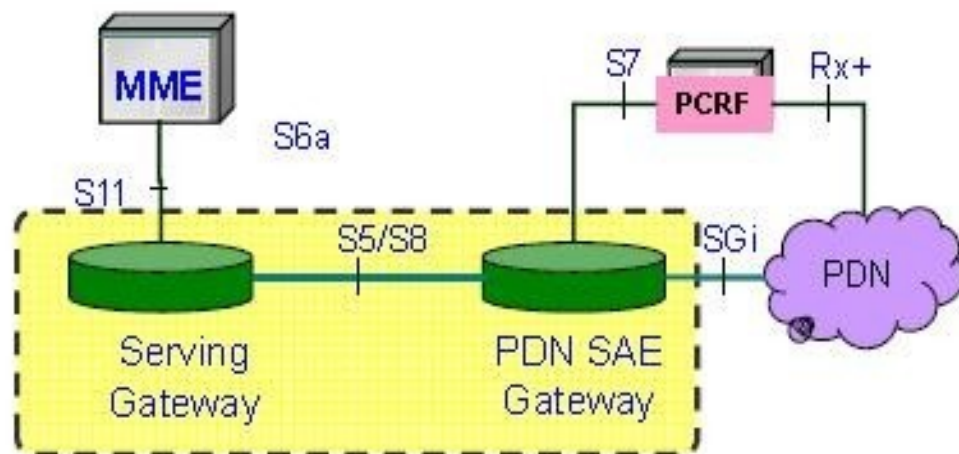
Packet Routing/Forwarding between Serving GW and external Data Network

Packet screening (firewall functionality)

- The PDN gateway provides the connection between EPC and a number of external data networks.
- Thus it is comparable to GGSN in 2G/3G networks.
- A major functionality provided by a PDN gateway is the QoS coordination between the external PDN and EPC.
- Therefore the PDN gateway can be connected via S7 to a PCRF (Policy and Charging Rule Function).

EPC (4-5)

Policy and Charging Rule Function (PCRF)



PCRF: Policy & Charging Rule Function

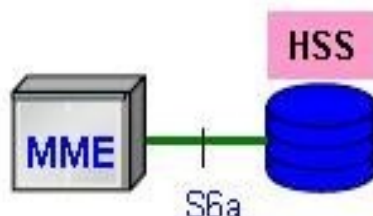
QoS policy negotiation with PDN

Charging Policy: determines how packets should be accounted

- The PCRF major functionality is the Quality of Service (QoS) coordination between the external PDN and EPC.
- Therefore the PCRF is connected via Rx+ interface to the external Data network (PDN)
- This function can be used to check and modify the QoS associated with a SAE bearer setup from SAE or to request the setup of a SAE bearer from the PDN.
- This QoS management resembles the policy and charging control framework introduced for IMS with UMTS release 6

EPC (5-5)

Home Subscriber Server (HSS)



HSS Functions

Permanent and central subscriber database

Stores mobility and service data for every subscriber

Contains the Authentication Center (AuC) functionality.

- The HSS is already introduced by UMTS release 5.
- With LTE/SAE the HSS will get additionally data per subscriber for SAE mobility and service handling.
- Some changes in the database as well as in the HSS protocol (DIAMETER) will be necessary to enable HSS for LTE/SAE.
- The HSS can be accessed by the MME via S6a interface.